

Surface Areas and Volumes (Set B) — MCQ Worksheet

Mathematics • Chapter 13 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Volume of a cube of side 4 cm is:

- (A) 64 cm^3
- (C) 48 cm^3

- (B) 16 cm^3
- (D) 96 cm^3

Q2. Total surface area of a cube of side 5 cm is:

- (A) 125 cm^2
- (C) 100 cm^2

- (B) 150 cm^2
- (D) 75 cm^2

Q3. Volume of cuboid $5 \times 4 \times 3$ cm:

- (A) 60 cm^3
- (C) 72 cm^3

- (B) 47 cm^3
- (D) 12 cm^3

Q4. Volume of cylinder radius 7 cm, height 10 cm ($\pi = 22/7$):

- (A) 1540 cm^3
- (C) 154 cm^3

- (B) 440 cm^3
- (D) 1100 cm^3

Q5. Curved surface area of cylinder radius 7 cm, height 10 cm ($\pi = 22/7$):

- (A) 440 cm^2
- (C) 880 cm^2

- (B) 154 cm^2
- (D) 220 cm^2

Q6. Volume of sphere of radius 21 cm ($\pi = 22/7$):

- (A) 38808 cm^3
- (C) 1386 cm^3

- (B) 9702 cm^3
- (D) 5544 cm^3

Q7. Volume of hemisphere of radius 7 cm ($\pi = 22/7$):

- (A) 718.67 cm^3
- (C) 342.67 cm^3

- (B) 1437.33 cm^3
- (D) 77 cm^3

Q8. Slant height of cone of radius 5 cm, height 12 cm:

- (A) 13 cm
- (C) 169 cm

- (B) 17 cm
- (D) 8.5 cm

Q9. Curved surface area of cone of radius 7 cm, slant height 25 cm ($\pi = 22/7$):

- (A) 550 cm^2
- (C) 275 cm^2

- (B) 154 cm^2
- (D) 1100 cm^2

Q10. If volume of a cube is 1000 cm^3 , side length is:

- (A) 10 cm
- (C) 1000 cm

- (B) 100 cm
- (D) 33.3 cm

Q11. Total surface area of a sphere of radius 14 cm ($\pi = 22/7$):

- (A) 2464 cm^2
- (C) 1232 cm^2

- (B) 616 cm^2
- (D) 770 cm^2

Q12. Volume of cone radius 6 cm, height 7 cm ($\pi = 22/7$):

- (A) 264 cm^3
- (C) 132 cm^3

- (B) 88 cm^3
- (D) 528 cm^3

Q13. Volume of a hemispherical bowl of radius 21 cm ($\pi = 22/7$):

- (A) 19404 cm^3
- (C) 1386 cm^3

- (B) 9702 cm^3
- (D) 5544 cm^3

Q14. If diameter of a sphere is doubled, its volume becomes:

- (A) 2 times
- (C) 8 times

- (B) 4 times
- (D) 16 times

- Q15.** A solid metallic ball of radius 6 cm is melted and recast into 27 small balls of equal radius. Radius of each small ball:
- (A) 2 cm (B) 3 cm
(C) 1 cm (D) 6 cm
- Q16.** A cone, hemisphere and cylinder stand on equal bases and have equal heights. Their volumes are in ratio:
- (A) 1:2:3 (B) 3:2:1
(C) 2:3:1 (D) 1:1:1
- Q17.** A sphere of radius 10 cm is melted and cast into wires of length 8 cm and radius 1 cm. Number of wires:
- (A) About 167 (B) 200
(C) 150 (D) 100
- Q18.** A solid right circular cone is cut into 2 parts at midpoint of axis by a plane parallel to base. The ratio of the volumes of two parts is:
- (A) 1:7 (B) 1:8
(C) 1:3 (D) 1:4
- Q19.** A hemispherical bowl has internal diameter 36 cm. The volume of liquid it can hold ($\pi = 22/7$):
- (A) 12235.7 cm³ (B) 6118 cm³
(C) 9237 cm³ (D) 4860 cm³
- Q20.** A right circular cylindrical pipe is 14 m long. Its inside and outside diameters are 16 cm and 20 cm. The volume of metal used ($\pi = 22/7$):
- (A) 158400 cm³ (B) 176000 cm³
(C) 317800 cm³ (D) 880000 cm³
- Q21.** A toy is in the form of a cone surmounted on a hemisphere with same radius 3.5 cm. Total height of toy = 15.5 cm. Total surface area is ($\pi = 22/7$):
- (A) 214.5 cm² (B) 77 cm²
(C) 100 cm² (D) 150 cm²
- Q22.** A solid sphere of radius r is melted and recast into a cone of base radius r and height h. Find h.
- (A) 4r (B) 3r
(C) 2r (D) r
- Q23.** Three solid spheres of radii 3 cm, 4 cm and 5 cm are melted and recast into a single solid sphere. Radius of new sphere:
- (A) 6 cm (B) 5 cm
(C) 4 cm (D) 7 cm
- Q24.** A cylindrical vessel of height 14 cm and diameter 40 cm is filled with sand. The sand is poured to form a conical heap of height 12 cm. Find the radius of conical heap.
- (A) $20\sqrt{(35/12)}$ cm (B) 20 cm
(C) 12 cm (D) Cannot determine
- Q25.** An iron pillar consists of a cylindrical portion 2.8 m high and 20 cm radius and a cone 42 cm high mounted on it. Weight of pillar at 7.5 g/cm³ ($\pi = 22/7$):
- (A) About 2738 kg (B) 1000 kg
(C) 5000 kg (D) 300 kg